

```
// math.exp (f16, max_bits_lost = 9) --lower-exp-to-polynomial-->

%r = polynomial.eval #polynomial.typed_chebyshev_polynomial<[0.9355, 0.4324, 0.1998]> : !polynomial.polynomial<ring = <coefficientType = f16>>, %x {scheme = "clenshaw", domain_lower = -2.0 : f16, domain_upper = 0.0 : f16} : f16

// --expand-polynomial-eval--> 12 branch-free mulf/addf/subf ops (no branches, no memory)
```

 approximation family (Chebyshev basis; extensible)

 approximation domain $[lower, upper]$ $\Rightarrow$ affine map to $[-1, 1]$

 coefficients $c_0 \dots c_d \Rightarrow$ degree d (DCT-I, compile time)

 evaluation scheme (Clenshaw recurrence; extensible)